

Question	Answer
1a	$\frac{2x+1}{2} \geq \frac{5-4x}{3}$
	$3(2x+1) \geq 2(5-4x)$
	$6x+3 \geq 10-8x$
	$14x \geq 7$
	$x \geq \frac{1}{2}$
1b	$6x+5y=8$ Eqn 1
	$10x+3y=24$ Eqn 2
	Eqn 1 $\times$ 3 $18x+15y=24$ Eqn 3
	Eqn 2 $\times$ 5 $50x+15y=120$ Eqn 4
	Eqn 4 $-$ Eqn 3 $32x=96$
	$x=3$
	$y=-2$
1c	$\frac{3x}{(x-5)^2} + \frac{2}{5-x}$
	$= \frac{3x}{(x-5)^2} - \frac{2}{x-5}$ Alt: $= \frac{3x}{(5-x)^2} + \frac{2}{5-x}$
	$= \frac{3x-2(x-5)}{(x-5)^2}$ $= \frac{3x+2(5-x)}{(5-x)^2}$
	$= \frac{x+10}{(x-5)^2}$
1d	$\left(\frac{\sqrt[3]{p^2}}{\sqrt{2q}} \right)^{-6}$
	$= \left(\frac{\sqrt{2q}}{\sqrt[3]{p^2}} \right)^6$
	$= \frac{(2q)^3}{(p^2)^2}$
	$= \frac{8q^3}{p^4}$
1e	$\frac{36x^2-4}{9x^2-6x+1}$
	$= \frac{4(3x-1)(3x+1)}{(3x-1)^2}$
	$= \frac{4(3x+1)}{3x-1}$

Question	Answer
2a	$\frac{0.75 \times 6 + 1.25 \times 14 + 1.75 \times x + 2.25 \times 10 + 2.75 \times 4}{6 + 14 + x + 10 + 4} = 1.67$
	$1.75x + 55.5 = 1.67(x + 34)$
	$1.75x + 55.5 = 1.67x + 56.78$
	$0.08x = 1.28$
	$x = 16$
2b	0.56
2c	The durians on Farm X are heavier because Farm X's mean of 1.67 kg is larger than Farm Y's mean of 1.45 kg.
2d	The standard deviation becomes smaller because the additional values are closer to the mean and so the total difference from mean is reduced.

Question	Answer
3a	2.92
3b	
3c	Well-balanced tangent line drawn 1.5 ± 0.3
3d.i	$x^3 + 4x^2 + 12x - 8 = 0$
	$x^2 + 4x + 12 - \frac{8}{x} = 0$
	$x^2 - \frac{8}{x} = -4x - 12$
	$\frac{x^2}{4} - \frac{2}{x} = -x - 3$
	$a = -1; b = -3$
3d.ii	Line $y = -x - 3$ correctly drawn
	$x = 0.552 \pm 0.05$
3e	1.9 ± 0.1

Question	Answer
4a.i	$\angle CAP = \angle DBQ = 90^\circ$ (tangent perpendicular to radius)
	$\angle ACP = \angle BDQ$ (base angle of isosceles triangle, $OC=OD$)
	$AC = BD$ (diameters of a circle)
	$\triangle APC$ is congruent to $\triangle BQD$ (AAS congruence test)
4a.ii	$\angle ACD = 180^\circ - 90^\circ - 54^\circ = 36^\circ$ (sum of angles in a triangle)
	$\angle ABD = 36^\circ$ (angles in the same segment)
4b	Reflex angle $XOY = 2\pi - 1.8$
	Area of major sector = $\frac{1}{2}(10)^2(2\pi - 1.8)$
	= 224.15 cm
	Area of triangle = $\frac{1}{2}(10)(10)\sin(1.8)$
	= 48.492 cm
	Area of unshaded section = $224.15 + 48.492$
	= 273 cm ²

Question	Answer
5a	$\frac{\sin \angle ACB}{17} = \frac{\sin 25^\circ}{9}$
	$\sin \angle ACB = \frac{\sin 25^\circ}{9} \times 17$
	$\angle ACB = \sin^{-1} \left(\frac{\sin 25^\circ}{9} \times 17 \right)$
	$= 52.966^\circ \text{ or } 127.034^\circ$
	$= 127.0^\circ$
5b	Angle $BAC = 180^\circ - 25^\circ - 127.0^\circ = 28.0^\circ$
	Area of triangle $ABC = \frac{1}{2} \times 17 \times 9 \times \sin 28.0^\circ$
	$= 25.914 \text{ cm}^2$
	Volume $= 25.914 \times 14$
	$= 502.796 = 503 \text{ cm}^3$
5c	$\frac{BC}{\sin 28.0^\circ} = \frac{9}{\sin 25^\circ}$
	$BC = \frac{9}{\sin 25^\circ} \times \sin 28.0^\circ$
	$= 9.9977$
	$BF = \sqrt{14^2 + 9.9977^2}$
	$= 17.203$
	$BD = \sqrt{14^2 + 17^2}$
	$= 22.022$
	$22.022^2 = 9^2 + 17.203^2 - 2(9)(17.203)\cos \angle DBF$
	$\cos \angle DBF = \frac{9^2 + 17.203^2 - 22.022^2}{2(9)(17.203)}$
	$\angle DBF = 110.4^\circ$

Question	Answer
6a	$\frac{x}{5} = \frac{13.5}{10+5}$
	$x = \frac{13.5}{15} \times 5$
	$x = 4.5$
6b.i	The volume of water is not linearly related to its height.
6b.ii	Height of empty space = 10 cm
	$\frac{V_1}{V_2} = \left(\frac{10}{15}\right)^3 = \frac{8}{27}$
	$\frac{\text{Volume of water}}{\text{Volume of cone}} = \frac{19}{27}$
	$\text{Volume of cone} = \frac{1}{3} \times \pi \times \left(\frac{13.5}{2}\right)^2 \times (15)$
	$= \frac{3645\pi}{16}$
	$\text{Volume of water} = \frac{19}{27} \times \frac{3645\pi}{16}$
	$= 503.63 = 504 \text{ cm}^3$
6b.iii	$\text{Area} = \frac{503.63}{7}$
	$= 71.947 \text{ cm}^3$
	$\text{Length} = \sqrt{71.947}$
	$= 8.48 \text{ cm}^3$

Question	Answer
7a	$\begin{pmatrix} 30 & 7 & 5 \\ 50 & 12 & 8 \end{pmatrix}$
7b	$\begin{pmatrix} 30 & 7 & 5 \\ 50 & 12 & 8 \end{pmatrix} \begin{pmatrix} 80 & 110 \\ -30 & -40 \\ -20 & -25 \end{pmatrix}$
	$= \begin{pmatrix} 30 \times 80 + 7 \times (-30) + 5 \times (-20) & 30 \times 110 + 7 \times (-40) + 5 \times (-25) \\ 50 \times 80 + 12 \times (-30) + 8 \times (-20) & 50 \times 110 + 12 \times (-40) + 8 \times (-25) \end{pmatrix}$
	$= \begin{pmatrix} 2090 & 2895 \\ 3480 & 4820 \end{pmatrix}$
7c	2090 represents the total ticket price of Group A going to the theme park on a weekday.
7d	$(3480 + 4820) \times 0.9$
	$= \$7470$
7e	$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$

Question	Answer
8a	$\sqrt{4^2 + 2^2}$
	$= 4.47$
8b	$\overrightarrow{SR} = \overrightarrow{PQ} = \begin{pmatrix} -4 \\ 2 \end{pmatrix}$
	Gradient of $SR = \frac{2}{-4} = -\frac{1}{2}$
	$y - 7 = -\frac{1}{2}(x - 2)$
	$y = -\frac{1}{2}x + 8$
	Alt for first 2 marks: Find point $S = (6, 5)$ by counting. M1 Find gradient from $(2, 7)$ and $(6, 5)$ M1
8c.i	Since all the angles in the rectangle are right angles, by the property of right angle in a semi-circle, both PR and QS are the diameters of the circle.
8c.ii	$\overrightarrow{PR} = \overrightarrow{PO} + \overrightarrow{OR}$
	$\overrightarrow{PR} = \begin{pmatrix} -3 \\ 1 \end{pmatrix} + \begin{pmatrix} 2 \\ 7 \end{pmatrix} = \begin{pmatrix} -1 \\ 8 \end{pmatrix}$
	By counting, $\overrightarrow{PC} = \begin{pmatrix} -0.5 \\ 4 \end{pmatrix}$ or $\overrightarrow{PC} = \frac{1}{2} \begin{pmatrix} -1 \\ 8 \end{pmatrix} = \begin{pmatrix} -0.5 \\ 4 \end{pmatrix}$
8c.iii	$\overrightarrow{OC} = \overrightarrow{OP} + \overrightarrow{PC} = \begin{pmatrix} 3 \\ -1 \end{pmatrix} + \begin{pmatrix} -0.5 \\ 4 \end{pmatrix}$
	$\overrightarrow{OC} = \begin{pmatrix} 2.5 \\ 3 \end{pmatrix}$

Question	Answer
9a	$\frac{1}{x+8}$
9b	$\frac{7}{x+8} \times \frac{6}{x+7} + \frac{x}{x+8} \times \frac{x-1}{x+7} = \frac{2}{5}$
	$\frac{42+x^2-x}{(x+8)(x+7)} = \frac{2}{5}$
	$5(42+x^2-x) = 2(x^2+15x+56)$
	$5x^2-5x+210-2x^2-30x-112=0$
	$3x^2-35x+98=0$
9c	$x = \frac{-(-35) \pm \sqrt{(-35)^2 - 4(3)(98)}}{2(3)}$
	$x = 7 \text{ or } 4\frac{2}{3}$
9d	$\frac{7}{15} \times \frac{1}{14} + \frac{1}{15} \times \frac{7}{14}$
	$= \frac{1}{15}$

Question	Answer
10a	$TDEE = [10 \times 72 + 6.25 \times 172 - 5 \times 25 + 5] \times 1.725$ $= 2889.375 \text{ calories/day}$
10b	$Calories = 7.3 \times 65 \times 0.75$
	$= 355.875 \text{ calories}$
10c	Weekly calories eaten $= 2150 \times 7 = 15050$
	$TDEE = [10 \times 75 + 6.25 \times 168 - 5 \times 42 + 5] \times 1.55$ $= 2472.25$
	Weekly difference $= 15050 - (2472.25 \times 7) = -2255.75$
	Calories spent in swimming $= 7.8 \times 75 \times 1 = 585$
	Calories for jogging $= 3850 - 2255.75 - 585 = 1009.25$
	Jogging time $= \frac{1009.25}{75 \times 8.8} = 1.5291 \text{ hr}$
	Each session $= \frac{1.5291}{3} \times 60 = 30.583 \approx 35$